Institute for Informatics
Georg-August-Universität Göttingen

Databases and Information Systems

LoPiX Logic Programming in XML



The LoPiX-Mondial Case Study: Data Integration in XML

The LoPiX - Mondial case study illustrates data integration in XML using the LoPiX system. LoPiX is an implementation of the XML XML querying and data manipulation language XPathLog.

The case study is based on XML representations of the sources which have been used for generating the Mondial database:

- CIA World Factbook,
- Global Statistics which has been collected by Johan van der Heijden.
- additional textual sources for coordinates,
- the *International Atlas* by Kümmerly & Frey, Rand McNally, and Westermann,
- and some geographical data of the Karlsruhe TERRA database.

The original HTML sources have been mapped to XML using wrappers with the FLORID system (see here).

- **The individual sources in XML format**
  - CIA Factbook: Country Data cia.xml cia.dtd
  - CIA Factbook: Organizations orgs.xml orgs.dtd
  - Global Statistics: Countries, Cities and Provinces gs.xml gs.dtd
  - Cities: Geographic Coordinates qiblih.xml qiblih.dtd
  - the Original TERRA database terra.xml terra.dtd
  - Country codes and names codes.xml codes.dtd

  (note that most browsers to not *show* XML and DTD files correctly. Download the file(s) and load them into an editor)

The integration of the above sources has been done in XPathLog:

- The integration is described in the in the report
  Information Integration in XML: The Mondial Case Study.
  Wolfgang May; Preliminary Report, Institut für Informatik, Universität Freiburg, March 2001.
- **Integration Program**
  - Integration program "mondial-integra.lpx"
- **Integrated XML Database**
  - The XML database can be found at the Mondial homepage.
    (Note that in since then, the database has been revised).